Highlights

Speech Large Language Models in the Real World: A Survey of Training, Evaluation, Deployment, and Responsible Design

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- Frames Speech LLMs as coupled real-world systems spanning training, evaluation, deployment, and responsible design.
- Synthesizes training strategies for speech–text alignment, weak supervision, adaptation, and preference alignment.
- Meta-evaluates recent Speech LLM benchmarks through construct, ecological, evaluator, and operational validity.
- Reviews deployment constraints including streaming inference, serving, edge feasibility, efficiency, and full-duplex interaction.
- Develops a speech-specific safety and governance framework for audio jailbreaks, voice cloning, privacy, and provenance.

Speech Large Language Models in the Real World: A Survey of Training, Evaluation, Deployment, and Responsible Design

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ABSTRACT

Speech Large Language Models (Speech LLMs) extend text-based language models to spoken interaction, preserving linguistic content together with paralinguistic cues such as emotion, prosody, and speaker identity. Existing surveys describe how Speech LLMs are built; this survey instead asks what evidence and system conditions are required before Speech LLMs can be evaluated, deployed, and governed as real-world interactive systems. We adopt a *lifecycle* framing connecting four stages—training, evaluation, deployment, and responsible design—and argue that progress in any one stage is constrained by choices in the others. The survey makes four contributions. First, we synthesize training strategies for modality alignment, weakly supervised pretraining, parameter-efficient fine-tuning, and preference alignment via RLHF, DPO, and RLAIF. Second, we conduct a meta-evaluation of recent Speech LLM benchmarks, analyzing construct, ecological, evaluator, and operational validity rather than only capability coverage. Third, we synthesize deployment research around real-time interaction constraints—streaming, serving, edge feasibility, and full-duplex behavior. Fourth, we develop a speech-specific responsible-design framework covering audio jailbreaks, voice cloning, speaker privacy, and provenance. By treating Speech LLMs as interactive systems rather than isolated models, the survey identifies the structural couplings that future research must address.

1. Introduction

Speech is the most natural channel of human communication, and Speech Large Language Models (Speech LLMs) are now closing the gap between text-based language modeling and real-time spoken interaction [16, 62]. As consumer voice assistants, agentic voice systems, and full-duplex models move from benchmarks into production, the question is no longer only how to train a Speech LLM, but whether the trained model can be evaluated reliably, deployed responsively, and governed responsibly.

Why text-LLM recipes do not transfer. A cascaded Automatic Speech Recognition (ASR) → LLM → Text-to-Speech (TTS) pipeline is the historical default, but it discards paralinguistic content, accumulates latency across stages, and propagates errors downstream [2, 20, 23]. End-to-end Speech LLMs remove these bottlenecks in principle, but the speech modality introduces challenges that text-only pipelines do not face: parallel speech-text data is scarce [19, 96], modality gaps between audio and text are non-trivial to bridge [23], and paralinguistic signals carry meaning that purely lexical metrics cannot capture [116].

Lifecycle framing as sharpening, not broadening. A natural reaction to these challenges is to extend an existing Speech LLM survey by appending evaluation, deployment, and safety as parallel chapters. We argue this is insufficient because the four stages are *coupled*: training choices determine what evaluation can reveal; evaluation gaps determine which deployment failures slip through; deployment constraints reshape acceptable training trade-offs; and safety risks span every stage. Rather than broadening the topic, the lifecycle framing sharpens it—it forces every section to be read through the lens of what evidence is missing for the others. This positioning distinguishes our work from prior Speech LLM surveys [20, 81] and broader audio-language reviews, which catalog how models are built but do not audit the conditions under which they can be trusted in deployment.

Contributions. This survey makes four contributions.

1. **Lifecycle framing.** We propose a lifecycle treatment that connects training, evaluation, deployment, and responsible design, and we make the coupling concrete through Figure 2.

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Table 1
Concrete cross-stage couplings observed in the literature covered by this survey. Each row links a specific upstream decision to a documented downstream consequence, illustrating that the four lifecycle stages are not independent.

Upstream stage	Decision / observation	Downstream stage affected	Documented consequence
Training	Whisper-style weakly supervised pretraining emphasizes transcription quality [83].	Evaluation	Benchmarks inherit ASR-centric metrics; paralinguistic and prosodic validity are under-measured (Section 4.4).
Training	Adapter / Q-Former bridge between speech encoder and language model [24, 97].	Deployment	Bridge layer adds latency; full-duplex stacks must redesign streaming boundary (Section 5).
Evaluation	Multilinguality coverage is fragmented across incompatible metrics [59, 79].	Deployment	Practitioners cannot rank multilingual systems for production without re-running on a target language slice (Section 5).
Evaluation	LLM-as-judge protocols underlie open-ended scoring with only moderate human correlation [5, 121].	Safety	Judge-based reward signals can miss unsafe prosody or jailbreaks triggered by acoustic content (Section 6).
Deployment	Quantization and KV-cache pruning are applied for edge feasibility [105].	Safety	Downstream safety classifiers and watermark detectors can degrade under aggressive compression (Section 6).

2. **Meta-evaluation of benchmarks.** In Section 4 we audit recent Speech LLM benchmarks not just for capability coverage but for construct, ecological, evaluator, and operational validity, identifying structural blind spots in what current benchmarks measure.
3. **Deployment synthesis around interaction constraints.** In Section 5 we synthesize deployment research around streaming inference, serving systems, edge feasibility, compression, and full-duplex behavior, rather than treating efficiency as a stand-alone engineering concern.
4. **Speech-specific responsible-design framework.** In Section 6 we develop a governance treatment that takes seriously what text-only safeguards miss in audio: jailbreaks via spoken input, voice cloning, speaker re-identification, and provenance watermarking.

Reading guide. Section 2 introduces the speech-specific concepts the survey relies on. Section 3 consolidates training strategies. Sections 4, 5, and 6 develop the lifecycle audit for evaluation, deployment, and safety. We close with open challenges (Section 7) and a synthesis (Section 8). The lifecycle map in Figure 2 traces these stages and the couplings the survey emphasizes.

To make the cross-stage couplings concrete, Table 1 records five examples where a training, evaluation, or deployment choice documented in the literature produced a visible consequence in a downstream stage.

2. Background: Speech LLMs and Their Lifecycle Challenges

We assume reader familiarity with the Transformer-based language model backbone (e.g., [11, 84, 100]) and standard supervised, instruction-tuned, and RLHF-aligned text LLM pipelines. This section sets up the speech-specific concepts the rest of the survey relies on.

Speech LLMs. We use *Speech LLM* to refer to an LLM-based system that consumes, reasons over, or generates speech as part of language interaction. This includes speech-to-text recognition coupled with an LLM (cascaded), speech-to-text and speech-to-speech models that share a single backbone (end-to-end), and omni-modal models that integrate speech with text and other modalities [20, 81].

Why speech is different. Three properties of speech shape every lifecycle stage of Speech LLMs. First, speech is a continuous, high-dimensional, multi-channel signal that carries linguistic content, paralinguistics, speaker identity, scene context, and timing simultaneously. Second, parallel speech-text data is expensive and unevenly distributed across languages and domains, so most training pipelines depend on weak supervision or self-supervision [36, 83].

Third, speech interaction is inherently real-time: a model that produces a correct transcript or response too late may fail the interaction even if its content is accurate. Together, these properties drive the design choices reviewed in the following sections.

Lifecycle framing. The remainder of the survey is organized around four lifecycle stages: training strategies (Section 3), evaluation and benchmarking (Section 4), deployment and real-time interaction (Section 5), and safety, privacy, and governance (Section 6). The four stages are tightly coupled. Training choices determine what benchmarks reveal; benchmarks reveal what deployment must defend against; deployment constraints reshape which training and evaluation trade-offs are acceptable; and safety risks span all three.

3. Training Strategies for Speech LLMs

Building Speech LLMs requires extending the text-LLM training pipeline to account for the continuous, multi-channel nature of speech. We organize the literature around four training stages: (i) modality alignment and pretraining, (ii) fine-tuning and adaptation, (iii) preference alignment, and (iv) cross-cutting design issues that connect training to the remainder of the lifecycle. Throughout, we emphasize choices that have downstream implications for evaluation (Section 4), deployment (Section 5), and safety (Section 6).

3.1. Modality Alignment and Pretraining

Speech and text differ in two ways that complicate joint modeling: speech sequences are typically much longer than their text counterparts, and speech embeddings live in a separate representation space from text embeddings [8, 24]. Modality alignment modules and acoustic pretraining strategies address these mismatches.

Alignment modules. Modular Speech LLMs combine a pretrained speech encoder (e.g., Whisper [83], HuBERT [36], wav2vec 2.0 [6]) with a partly frozen text LLM through a trainable bridge. *AlignFormer* [24] pairs a Connectionist Temporal Classification (CTC) layer with a Q-Former-style attention module to map variable-length acoustic features to discrete text-aligned tokens. *Seed-ASR Converter* [8] learns a context-aware transformation that improves multilingual generalization. *SALMONN*'s Q-Former [97] and the OSUM adapter [31] use a small set of trainable queries to distill semantically rich acoustic representations into the LLM's input space. As an alternative to adapter-based bridging, X-OPD [12] distills text-LLM capabilities into a speech-LLM via cross-modal on-policy rollouts, bypassing the need for paired speech-text supervision altogether. Empirical alignment quality can be probed via the Automatic Latent Alignment Score (ALAS) [70], which uses Whisper timestamps as a reference and reports that deeper transformer layers align better for semantic tasks while earlier layers align better for paralinguistic tasks.

Self-supervised pretraining. Speech LLMs adapt the masked-prediction and clustering objectives from text SSL to raw audio. Representative methods include UniSpeech-SAT [14], which adds speaker-aware objectives to the SSL loss; DinoSR [58], which combines self-distillation with masked acoustic modeling; and Seed-ASR [8], which scales a BERT-style masked-prediction pretraining to millions of hours of speech to produce the LUISE encoder, subsequently aligned to a language model.

Joint and weakly supervised pretraining. Two further pretraining paradigms reduce reliance on parallel speech-text data. *Cold-start joint modeling* initializes both speech and language components together, as in Generative Spoken Language Modeling (GSLM) [50] and SPIRIT-LM [72], yielding tighter cross-modal interactions than methods that freeze the language model. *Weakly supervised pretraining at scale*, exemplified by Whisper [83]'s 680,000-hour multilingual corpus, trades exact transcription quality for breadth and produces strong zero-shot transcription, translation, and detection capabilities. Many later systems (OSUM [31], SALMONN [97]) build directly on Whisper-style encoders, treating large-scale weak supervision as the de facto foundation.

3.2. Fine-Tuning and Adaptation

Pretraining yields foundations; fine-tuning produces task-aligned behavior. Speech LLM fine-tuning takes three main forms.

Instruction tuning. Instruction tuning aligns Speech LLMs to user intent in spoken dialogue, question answering, and editing tasks. SpeechGPT [116] integrates discrete speech tokens into a decoder-only LLM backbone with strong zero-shot generalization. COSMIC [78] trains a Q-Former bridge on 450 hours of speech plus GPT-3.5-generated QA pairs, achieving competitive ASR and speech-to-text translation with limited paired data. InstructSpeech [38] uses hierarchical adapters and multi-step reasoning for fine-grained semantic and acoustic editing. LLaMA-Omni [25] pairs a frozen Whisper encoder with LLaMA-3.1-8B-Instruct and a streaming decoder, reporting 226 ms response latency without intermediate transcription. More recently, Audio-FLAN [109] demonstrates that scaling instruction data to over 100 million instances across 80 audio tasks yields substantial gains, signaling a shift toward massive-scale instruction tuning.

Parameter-efficient tuning. Parameter-efficient fine-tuning (PEFT) updates a small fraction of model parameters while freezing the rest, reducing compute cost and mitigating catastrophic forgetting. HDMoLE [71] combines hierarchical LoRA adapters with mixture-of-experts routing for joint accent and domain adaptation, updating ∼9.6% of parameters while matching full fine-tuning. ELP-Adapters [39] insert lightweight modules at multiple abstraction levels of pretrained encoders such as WavLM, enabling targeted adaptation with minimal parameter footprint. Zipper-LoRA [67] further advances speech-specific PEFT by introducing language-conditioned routing with rank-level decoupling for multilingual speech recognition.

Domain-specific tuning. Specialized domains (medical, financial, technical) often have constrained paired speech-text resources. Low-resource adaptation via unpaired domain-specific text [26] fine-tunes the language decoder with LoRA while preserving alignment via real-time evaluation, reducing domain WER without harming general performance. Decoder-only context perturbation [95] freezes the encoder and randomizes domain descriptions during decoder fine-tuning to mitigate overfitting and forgetting, with successful applications to earnings calls and medical dialogue.

3.3. Post-Training Preference Alignment

Pretraining and fine-tuning equip Speech LLMs with capability; preference alignment shapes their behavior to match user expectations for helpfulness, safety, and naturalness. This is especially consequential in speech, where prosody, pacing, and emotion strongly influence perceived quality [19].

Reinforcement Learning from Human Feedback (RLHF). RLHF [18, 76] trains a reward model on human-labeled preference pairs and then optimizes the base model against it. The reward model uses a pairwise ranking loss that converts human comparisons into a scalar reward:

$$\mathcal{L}^{\text{reward}}(\phi) = -\log \sigma\left(r_\phi(x, y_1) - r_\phi(x, y_2)\right), \tag{1}$$

where x is the prompt, y_1 is preferred over y_2, and r_ϕ scores responses. The base model is then optimized via policy-gradient methods (typically PPO [88] in text settings) with a KL constraint to a reference model. RLHF is expensive in human annotation and unstable to tune, particularly in speech, where preference data must capture acoustic and paralinguistic quality in addition to content. As a result, Speech LLMs typically substitute RLHF with lighter alternatives [85].

Direct Preference Optimization (DPO). DPO [85] eliminates the explicit reward model by expressing the preference loss directly in terms of the policy:

$$\mathcal{L}^{\text{DPO}}(\theta) = -\mathbb{E}_{x,y_w,y_l}\left[\log \sigma\left(\beta\left(\log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)\right)\right]. \tag{2}$$

DPO matches or exceeds RLHF on text alignment tasks while using a single training stage. Speech LLMs have rapidly adopted it. SpeechAlign [117] iteratively refines a zero-shot TTS model on LibriSpeech and VCTK, surpassing RLHF-PPO on Word Error Rate, Speaker Similarity, and human ratings. Align-SLM [55] extends DPO to a textless spoken language model trained on LibriSpeech and Multilingual LibriSpeech, outperforming Moshi and SPIRIT-LM on the spoken StoryCloze benchmark. OpenOmni [62] introduces a Connectionist Temporal Classification variant of DPO (CTC-DPO) for emotionally coherent speech generation, reporting strong results on the EO2S-9k bilingual emotional speech benchmark. Beyond pairwise comparisons, Emo-LiPO [57] proposes listwise preference optimization for speech emotion intensity, suggesting that richer ranking signals can further improve alignment quality.

Table 2
Comparison of preference alignment methods for Speech LLMs. The last column links each method to evaluation considerations developed in Section 4.

Aspect	RLHF	DPO	RLAIF	Speech-evaluation hook
Annotation source	Human preference pairs	Human preference pairs; some implementations mix synthetic data	AI-generated preference pairs	Speech requires acoustic + paralinguistic judgements; LLM judges are still moderate-correlation [121]
Optimization	Policy gradient (e.g., PPO)	Direct preference loss	Reward learning or DPO-style	Construct/evaluator validity (Section 4) determines which optimization signal is reliable
Reward model	Required	Not required	Required (or implicit)	A reward trained on transcripts misses acoustic quality; speech reward models remain underdeveloped
Engineering overhead	High (multi-stage pipeline)	Moderate (single-stage)	Moderate (auto-labeling pipeline)	Operational validity hooks: alignment cost shapes deployment cycle (Section 5)
Speech adoption	Limited (manual effort, slow iteration)	Promising: SpeechAlign, Align-SLM, OpenOmni	Emerging: RLAIF-V, AlignSLM	Adoption tracks the availability of speech-aware judges and benchmarks
Scalability	Limited by annotation cost	Improved efficiency	High (AI-supervised)	Scaling depends on whether evaluation can detect over-optimization (Section 4)

Reinforcement Learning from AI Feedback (RLAIF). RLAIF [9, 52] replaces human preference annotation with AI-generated labels, trading a small accuracy cost for substantially improved scalability. Lee et al. show that direct RLAIF (d-RLAIF) prompts an AI labeler at training time and matches or exceeds RLHF on summarization and dialogue. RLAIF-V [115] extends this to multimodal models with deconfounded generation and atomic claim decomposition; the 12B variant reduces its overall hallucination rate by 32.4% through self-alignment, surpassing GPT-4V in trustworthiness. In speech, AlignSLM [55] uses semantic metrics on AI-generated speech continuations to construct preference data for DPO, achieving state-of-the-art results on spoken StoryCloze.

Comparison. Table 2 compares RLHF, DPO, and RLAIF alongside their speech-specific evaluation considerations. The choice depends not only on annotation cost, but also on what evaluation regime (Section 4) is available: methods that require dense per-turn or per-utterance ratings are difficult when benchmarks can only score complete responses.

3.4. Cross-Cutting Training Issues

Three training-design decisions have outsized influence on the rest of the lifecycle and motivate later sections.

Data sufficiency. Self-supervised and weakly supervised pretraining on tens to hundreds of thousands of hours has become the default [8, 83], but parallel speech-text data remains scarce for low-resource languages, scripts, and domains. This data imbalance reappears in evaluation: most multilingual benchmarks under-sample non-Latin scripts (Section 4).

Alignment quality. Whether alignment modules preserve paralinguistic content varies substantially across architectures [24, 70]. Benchmarks such as SALMon and FMSU-Bench show that semantically capable Speech LLMs frequently fail at acoustic-semantic binding (Section 4), which suggests that alignment quality, not language modeling, is often the bottleneck.

Deployment-aware training. Few training pipelines optimize explicitly for streaming, latency, or interruption handling. Models designed for offline benchmarks may compute correctly but respond too late for natural interaction, motivating the architecture and protocol choices analyzed in Section 5. Safety-aware training is similarly underdeveloped: text safety alignment does not automatically transfer to audio inputs, as Section 6 demonstrates.

These cross-cutting issues connect the training section to the rest of the survey. The following sections examine whether current evaluation, deployment, and safety practices can measure the limitations introduced by training choices.

4. Evaluation and Benchmarking

This section is a meta-evaluation of Speech LLM benchmarks rather than a catalog. We ask three questions: what current benchmarks measure, how reliably they measure it, and what remains invisible to them. Building on text-LLM evaluation efforts such as HELM, MMLU, and BIG-Bench [34, 54, 93], we identify the speech-specific extensions that an IPM-quality survey must demand.

4.1. From LLM Evaluation to Speech LLM Evaluation

Evaluation is where claims about Speech LLM training become claims about capability, reliability, and deployability. Classical machine learning organizes evaluation around three elements: a dataset, a protocol, and a metric. LLM evaluation complicates all three. Open-ended generation weakens single-label scoring [54], prompt and decoding choices introduce non-trivial score variance [34], training-data contamination is difficult to rule out [93], and LLM-as-judge protocols create a circular dependency in which the evaluator itself becomes an object of evaluation [121].

Speech adds another layer of difficulty. A speech input is not a text sequence but a *multi-channel acoustic signal* carrying linguistic content, paralinguistic information, speaker traits, accent, acoustic scene, timing, and interaction dynamics simultaneously. Word error rate, BLEU, exact match, and MOS-style scores are therefore necessary but incomplete: each observes only a slice of speech behavior. A system can transcribe words correctly while missing emotion, produce a semantically plausible answer while ignoring background events, or generate a correct response too late for natural conversation. Recent benchmarks expose these failures from different angles:

- **Acoustic–semantic misalignment**: SALMon [64] probes whether models bind acoustic context to spoken content.
- **Fine-grained paralinguistic blindness**: FMSU-Bench [53] and VoxEmo [118] expose weaknesses in micro-acoustic perception.
- **Multilingual and code-switching fragility**: Dynamic-SUPERB [37] and SCENEBench [40] include explicit cross-lingual stress cases.
- **Interaction instability**: StyleBench [120] and VoiceAgentBench [41] target multi-turn behavior and tool use.
- **Cross-modal safety divergence**: VoiceBench [15] shows that safety responses can change between text and speech modes; we develop this further in Section 6.

The question for evaluation is therefore not only whether a model performs well on a task, but whether the benchmark measures the speech-specific behavior that matters for real use. Subsequent subsections audit whether current benchmarks meet that bar.

4.2. Benchmark Corpus and Scope

We analyze 18 benchmarks from 2024–2026 that evaluate speech-relevant capabilities of LLM-based systems. The pool includes benchmarks designed directly for Speech LLMs, broader Audio LLM benchmarks when their tasks transfer to speech interaction (judged by whether at least half their prompts contain spoken language as primary signal), and voice-assistant, agentic, multilingual, paralinguistic, and auditory-knowledge benchmarks when they evaluate speech-relevant behavior. Traditional ASR and TTS benchmarks are used only as background unless embedded in a Speech LLM evaluation protocol. Of the 18 benchmarks, 3 are peer-reviewed (SD-Eval at the NeurIPS 2024 Datasets and Benchmarks track [5], Dynamic-SUPERB at ICLR 2025 [37], SLUE-PERB at INTERSPEECH 2023 [89]); the remaining 15 are arXiv preprints as of June 2026 (we count only benchmarks with confirmed proceedings or journal publication, not papers under review). We label preprint evidence as such throughout and avoid claims that depend on un-peer-reviewed sample sizes or rankings.

For Figure 3, we coded capability coverage as a binary indicator: a dimension is counted when the benchmark explicitly defines a task, supplies test data, and reports a metric for that dimension; partial or implicit coverage was not counted. Disagreements during coding were resolved by re-reading the original paper rather than by inference.

Table 3
Primary placement of 18 Speech LLM benchmarks (2024–2026) under a scope × lens taxonomy. Several benchmarks have secondary lenses, indicated by footnote symbols. Scale = total samples (rounded); † = peer-reviewed; ⋆ = capability-focused secondary; ‡ = use-case-grounded secondary.

Scope / Lens	Capability	Use case	Phenomenon	Knowledge
Targeted probe	SALMon (8 tasks)[64]; VoxEmo (35 corpora)[118]; StyleBench (4 dims)[120]; SD-Eval† (4 dims, 7.3K)[5]	–	LoASR-Bench (25 langs)[13]; Hearing2Translate (16 benchmarks)[79]; SCENEBench (4 tasks)‡[40]	AKB-2000 (2K Q)[61]
Capability-focused	AudioBench (26 datasets, 100K+)[102]; AIR-Bench (19 tasks, 19K+)[111]; FMSU-Bench (14 dims, 20K)⋆[53]; SeaLLMs-Audio (1.58M train, 580 test)[59]	–	–	–
Holistic	Dynamic-SUPERB† (180 tasks)[37]; MMAU (10K Q, 27 skills)[87]	–	–	–
Use-case-grounded	–	VoiceBench (8 tasks, 3K)⋆[15]; VoiceAgentBench (6 tasks, 6K)[41]; Audio2Tool (8 tiers, 30K)[77]; SLUE-PERB† (4 tasks, 22K)[89]	–	–

Search and selection. Benchmarks were identified via Google Scholar, the arXiv `cs.CL` and `eess.AS` categories, the ACL Anthology, and the proceedings of NeurIPS, ICLR, ICML, INTERSPEECH, and ICASSP, covering January 2024 through June 2026. We searched for "Speech LLM", "Audio LLM", "voice assistant benchmark", "speech evaluation", "audio jailbreak", "full-duplex speech", and "streaming speech model". A benchmark was included when it evaluated an LLM-based system on speech or audio input, defined tasks with test data and metrics, and covered at least one capability dimension from Figure 3; pure ASR or TTS benchmarks were excluded unless they were embedded in a Speech LLM evaluation protocol. Coding for Figure 3 was performed by a single reviewer with disagreements resolved by re-reading the source paper rather than by inference; per-benchmark reading notes are released as supplementary material so that the coding can be independently validated by the community. We label this corpus as a 2024–2026 snapshot; given that fifteen of the eighteen benchmarks were arXiv preprints at the time of writing, some entries may have been peer-reviewed or superseded by the time of publication, and the analysis should be read as a contemporaneous audit rather than a final record.

4.3. A Two-Axis Taxonomy of Speech LLM Benchmarks

A single-axis taxonomy by task family is useful but insufficient. The same benchmark can be broad in modality coverage, narrow in evaluation lens, and partially anchored to a deployment use case. We organize the benchmark landscape using two orthogonal axes. The *scope* axis describes breadth and depth: targeted probe, capability-focused suite, holistic suite, or use-case-grounded evaluation. The *lens* axis describes the benchmark's dominant evaluative perspective: capability, use case, phenomenon, or knowledge. Table 3 shows primary placements with secondary lenses noted in footnotes; placements are not exclusive. VoiceBench is use-case-grounded but also probes robustness and safety; SCENEBench is primarily phenomenon-oriented but has use-case flavor; FMSU-Bench can be read as both capability-focused and a targeted paralinguistic probe.

We read Table 3 by row first (scope) and then by column (lens). Cells with multiple entries indicate saturation of attention; empty cells indicate evaluation blind spots.

The matrix exposes three structural gaps:

- **The empty Use-case × Capability cell.** No benchmark in our corpus of 18 asks whether a model can complete a realistic job *while* preserving a specific speech capability such as emotion awareness, code-switching fidelity, or robust tool calling under noise. A practitioner cannot select a Speech LLM for a real-time multilingual emotional assistant by reading benchmark scores.
- **Sparse Knowledge lens.** AKB-2000 [61] is the only benchmark in our corpus that systematically probes what auditory knowledge the LLM backbone encodes, despite backbone choice being a first-order Speech LLM design decision.

- **Narrow Phenomenon lens.** Multilinguality, translation, and acoustic-scene conditions are well represented (LoASR-Bench, Hearing2Translate, SCENEBench), but privacy, safety, latency, and interaction phenomena remain weakly integrated into mainstream benchmark design.

Beyond the structural placement of benchmarks, we now ask which capabilities they collectively measure.

4.4. What Current Benchmarks Actually Measure

The bar chart in Figure 3 summarizes 18 benchmarks across 17 capability dimensions (the full 18×17 binary matrix is released as supplementary material). We group capabilities into three tiers by frequency.

Saturated capabilities (≥10/18). ASR/transcription, speech QA or understanding, and emotion-related paralinguistics each appear in 12 of 18 benchmarks. This pattern reflects the field's dependence on inherited ASR and spoken-language-understanding tasks, plus emotion as the most accessible paralinguistic label.

Fragmented coverage (6–9/18). Audio-scene understanding, style or prosody, multilinguality, and speaker traits appear in nine benchmarks each; accent or dialect appears in seven; multi-turn dialogue appears in seven; noise and OOD robustness appear in six. These counts should not be read as maturity. Multilinguality, for example, appears in nine benchmarks but is measured incompatibly: SeaLLMs-Audio [59] covers five languages (Indonesian, Thai, Vietnamese, English, Chinese) with Gemini-judge scoring; LoASR-Bench [13] covers 25 languages with CER/WER; VoxEmo [118] covers 15 languages with distribution-aware soft labels. These three results are mutually incomparable.

Deployment-facing gaps (≤4/18). Safety appears in four benchmarks, code-switching in three, tool use in two, latency or streaming in only one (SCENEBench), speech generation quality in only one (StyleBench), and privacy or speaker anonymity in none of the 18 benchmarks analyzed. We elaborate the latency and safety gaps in Section 5 and Section 6, respectively.

The field is missing joint evaluation, not more benchmarks. A capability may appear in several benchmarks while still being measured under incompatible data sources, prompts, metrics, or judge protocols, and the dimensions most needed for deployment (latency, safety, privacy, generation quality) are precisely those least represented. We turn next to whether even the dimensions that *are* measured are measured reliably.

4.5. Methodological Validity: Data, Metrics, Judges, and Protocols

The central issue is validity: whether a benchmark measures what it claims to measure under conditions that support the conclusions being drawn. Four forms of validity are especially relevant.

Construct validity. Construct validity asks whether the scoring method matches the target capability. Multiple-choice formats, as in MMAU and AKB-2000 [61, 87], improve reproducibility but underrepresent open-ended semantic behavior. Rule-based free-response metrics such as WER, CER, F1, and ROUGE are objective but brittle under paraphrase, script variation, and downstream task semantics. LLM-as-judge scoring, used by AudioBench, VoiceBench, AIR-Bench, SD-Eval, and SeaLLMs-Audio [5, 15, 59, 102, 111], better supports open-ended responses but introduces judge choice, prompt sensitivity, and position-bias concerns formalized for text in [121]. SALMon [64] introduces a fifth paradigm—modeling-based pairwise preference via likelihood comparisons—that requires no external judge but is restricted to relative comparisons. For subjective targets such as emotion and style, single hard labels can be misleading because annotator disagreement is part of the phenomenon. VoxEmo's distribution-aware metrics [118] therefore offer a principled direction for paralinguistic evaluation, though the approach has so far been validated on only two models (Qwen2-Audio and Audio Flamingo). The field needs a second tier of metrics that match each capability's epistemic structure: distributional for ambiguous targets, modeling-based for relative comparison, and exact-match for pure information retrieval.

Ecological validity. Ecological validity asks whether benchmark data resembles real speech use. Real recordings preserve speaker variability, disfluency, and spontaneous acoustic artifacts, but they are costly and often have uneven demographic coverage. Synthetic TTS data is scalable and controllable, enabling benchmarks such as VoiceBench, StyleBench, VoiceAgentBench, and Audio2Tool [15, 41, 77, 120] to cover many prompts or scenarios; however, synthetic speech may smooth away idiosyncratic speaker behavior, fatigue, natural hesitation, and real environmental variation. Hybrid designs that overlay perturbations or scenes on real speech, as in SCENEBench and

Hearing2Translate [40, 79], are useful but still require validation against natural recordings. We therefore propose that synthetic-only benchmarks should not anchor deployment claims; any fairness, robustness, or deployment claim should include a real-speech validation slice.

Evaluator validity: judge agreement. Evaluator validity asks who or what judges the output. LLM judges make large-scale open-ended evaluation feasible, but reported agreement with humans is moderate rather than definitive. SD-Eval reports a Spearman correlation $\rho = 0.613$ between GPT-4o judge scores and human ratings [5]; SeaLLMs-Audio reports a Pearson correlation of 0.8 with 93% agreement excluding ties [59]; AIR-Bench explicitly documents position bias and mitigates with answer-swap voting [111]; the canonical text-LLM judge-bias literature [121] confirms that these issues are systemic, not benchmark-specific.

Evaluator validity: contamination risk. A more serious risk emerges when proprietary LLMs help generate benchmark items, filter annotations, or score outputs. The AKB-2000 paper [61] explicitly acknowledges that proprietary LLMs co-curated its questions, capping their performance at 94–96% as upper bound rather than independent measurement; FMSU-Bench [53] uses Gemini 2.5 Pro as primary annotator with multi-expert cross-validation; SeaLLMs-Audio uses Gemini-2.5-flash as judge for the same model class. In each case a model's score may partially reflect the curation pipeline rather than independent capability. Human baselines remain limited: only three benchmarks in our pool (MMAU, SD-Eval, SALMon) report human upper bounds at scale, making it hard to interpret whether a given score indicates saturation, progress, or metric artifact. We therefore propose that proprietary judges should not score the same model family that helped curate the benchmark, and that human baselines should accompany any new capability claim.

Operational validity. Operational validity asks whether the benchmark reflects deployment. Many benchmarks are valid for offline comparison but weak for choosing a real-time voice assistant. Latency, streaming mode, endpointing, audio chunking, hardware, batch size, interruption handling, and multi-turn stability are rarely standardized alongside task accuracy. SCENEBench [40] is the only benchmark in the pool that measures per-clip latency, and it does so in isolation from content quality. VoiceAgentBench [41] addresses tool use but explicitly does not study dynamic real-time tool invocation. This creates a deployment mismatch: benchmarks may report that a model understands speech while remaining silent about whether it can respond quickly, preserve paralinguistic cues, and remain safe during an interactive session. We elaborate the latency and streaming dimensions of this mismatch in Section 5. We therefore propose that latency, streaming, and content quality should be jointly reported, not separately.

Severity ranking. Among the four validity types, operational validity is the most damaging for IPM-relevant deployment claims: a benchmark can score perfectly on construct, ecological, and evaluator validity yet still fail to inform a practitioner choosing a real-time voice system. Construct and evaluator validity interact in direct tension—tightening one (e.g., requiring distribution-aware scoring) often weakens the other (e.g., by demanding LLM judges that cannot represent label distributions)—so they must be balanced rather than maximized independently.

4.6. Which Models Are Actually Evaluated? Reporting Bias and Deployment Mismatch

Existing benchmark surveys ask "what does each benchmark cover?" We instead ask "which models are systematically benchmarked, and which remain invisible?" The model-evaluation matrix concentrates around accessible or widely recognized systems. Qwen2-Audio [98] appears in nine benchmarks; Whisper variants [83], GPT-4o [74], and Whisper+LLM cascades each appear in seven; Audio Flamingo [48] and Qwen2.5-Omni [107] appear in six; SALMONN [97] and LLaMA variants [99] appear in five; pre-2.0 Qwen-Audio [7], Gemini variants [33], Qwen3-Omni [107], Mini-Omni [105], Kimi-Audio [22], and DeSTA2 [60] each appear in four. The concentration is understandable: open-weight models are easier to run, Whisper cascades serve as convenient baselines, and proprietary frontier models such as GPT-4o and Gemini provide recognizable reference points.

This concentration creates reporting bias. Coverage of interaction-first systems is sparse and asymmetric. Moshi [23] and the GPT-4o-Audio real-time mode each appear in a single benchmark (VoiceBench [15]). Four other interaction-first systems—MiniCPM-o 4.5, Voila, OmniFlatten, and Gemini Live [33]—do not appear in any benchmark in our pool. Ordinary GPT-4o or Gemini evaluations should be distinguished in practice from evaluations of their real-time audio interaction modes (the GPT-4o = 7 and Gemini = 4 counts in the matrix include text-mode evaluations; real-time-audio-mode evaluation is essentially absent from our pool). This is precisely the deployment regime that motivates the Speech-LLM concept, and it is precisely the regime that existing benchmarks fail to evaluate. We elaborate the deployment implications of this gap in Section 5.

4.7. Validity Gaps and Benchmark Design Principles

The four validity questions of Section 4.5 surface four parallel gaps. The construct validity gap: benchmarks repeatedly measure ASR, spoken QA, and emotion, but subjective and multi-channel capabilities need metrics that preserve ambiguity and cross-channel interaction. The ecological validity gap: synthetic data enables scale but weakens real-world claims unless paired with real-speech validation. The operational validity gap: latency, streaming, endpointing, interruption handling, and multi-turn stability are not jointly evaluated with content quality. The governance validity gap: safety, privacy, speaker anonymity, provenance, and identity risks are under-integrated into mainstream benchmark design and demand the extended treatment in Section 6.

We propose five design principles for future Speech LLM benchmarks:

1. **Multi-channel joint evaluation.** Each benchmark should report at least two of: content fidelity, paralinguistic awareness, interaction dynamics, latency, and safety. Audio2Tool's joint reporting of ASR, tool-use accuracy, and noise robustness [77] is a partial precedent.
2. **Distribution-aware scoring for subjective targets.** Emotion and style should use soft labels or distributional metrics when annotator disagreement is intrinsic, following VoxEmo's KLD/JSD/TVD template [118].
3. **Cross-source validation.** Claims based mainly on synthetic speech should be replicated on real-speech subsets. SCENEBench's pairing of 16k synthetic and 20 natural validation clips [40] illustrates the design pattern, though the natural-speech fraction remains negligibly small.
4. **Mandatory evaluation metadata.** Reports should document prompts, judge model and version, decoding settings, hardware, audio chunking, streaming mode, and whether any LLM was used during data curation.
5. **Speech-LLM Evaluation Cards.** Analogous to ML model cards [69], each benchmark should publish a structured card declaring: (a) primary and secondary lenses from Table 3; (b) capability cells from Figure 3 that the benchmark measures; (c) data source (real, synthetic, hybrid); (d) scoring method (rule-based, LLM-judge with model and version, distribution-aware, modeling-based); (e) human baseline availability; (f) model coverage including any $0/N$ omissions; and (g) acknowledged validity threats.

Of these, Principle 1 is the most immediately actionable: it requires no new dataset and can be retrofitted into current benchmark releases by adding latency and safety metrics alongside existing capability scores.

Evaluation is becoming a bottleneck for trustworthy Speech LLM deployment. Current benchmarks reveal individual capabilities but under-measure the interactions among speech content, timing, paralinguistics, safety, and real-world use that voice systems must handle simultaneously. The four validity gaps motivate the deployment-oriented analysis in Section 5 and the safety-and-governance analysis in Section 6.

5. Deployment: From Capable Models to Real-Time Systems

The preceding section showed that the evaluation of Speech LLMs must account for multi-channel capabilities and interaction behavior. Deployment introduces a complementary question: whether these capabilities can be delivered continuously under low latency, bounded memory, and realistic hardware constraints. A model that performs well on offline ASR, TTS, or speech understanding benchmarks may still fail as an interactive system if it cannot process streaming input, emit partial output, handle long acoustic contexts, or operate within the resource limits of the target device.

The release of GPT-4o in mid-2024 made this deployment gap broadly visible [74]. It showed that low-latency spoken interaction with multimodal models could approach human conversational timing under the reported conditions: users could speak naturally, receive spoken responses with sub-second latency, and experience an interaction loop that was more fluid than those typically supported by earlier cascaded systems. GPT-4o therefore provides a useful user-experience reference point for the deployment literature: subsequent open, edge, and streaming systems study how parts of this experience can be supported under more transparent or resource-constrained settings. However, reproducing this experience remains difficult outside large proprietary systems. The bottleneck is not only model capability; it is also deployment.

Speech is harder to deploy than text because it is a continuous, multi-channel signal. A deployed Speech LLM must often listen while speaking, preserve linguistic content while tracking paralinguistic cues, produce partial outputs before an utterance is complete, and remain responsive under variable acoustic conditions. These requirements change the serving problem. Batch throughput is no longer sufficient; first-byte latency, token delay, memory growth, turn-taking stability, and edge feasibility become first-order metrics. This section reviews recent work that addresses these constraints through serving systems, deployment-oriented architectures, streaming inference, algorithmic acceleration, and full-duplex omni-modal design.

5.1. The Deployment Predicament

Classical speech pipelines separated ASR, dialogue modeling, and TTS. This separation made deployment modular: each component could be optimized and monitored independently. Speech LLMs collapse this boundary. A single model, or a tightly coupled model stack, may now perform recognition, reasoning, response planning, and speech generation. This integration can improve interaction quality, but it also makes deployment failures harder to isolate.

Five recurring deployment failure modes motivate the analysis in this section. The first is latency fragmentation: papers report time to first byte (TTFB), time to first token (TTFT), average token delay, end-to-end latency, or throughput, but these metrics capture different parts of the interaction loop. The second is streaming mismatch: models trained or evaluated on complete utterances are often adapted to streaming use only after training. The third is memory growth: long audio and video streams create key-value cache (KV-cache) and feature-cache pressure that differs from text-only contexts. The fourth is hardware mismatch: theoretical FLOPs do not reliably predict performance on CPUs, mobile NPUs, or mixed edge-cloud deployments. The fifth is interaction incompleteness: many systems optimize recognition or generation latency in isolation, but they do not evaluate simultaneous listening and speaking.

These failure modes show that deployment cannot be summarized by a single latency or throughput number. It depends on how model architecture, decoding policy, memory management, serving infrastructure, and hardware interact under realistic speech workloads.

5.2. From Failure Modes to Deployment Analysis

Assessing deployment work requires decomposing each claim into three parts: the operational pressure the system addresses, the system response it applies, and the runtime conditions that produce the result. This structure turns latency, streaming, memory, hardware, and interaction issues into concrete review questions. Does the system reduce first-response delay, incremental token delay, or complete-response time? Does it support partial input during generation? How does memory grow with audio duration? Which device, runtime stack, and serving policy produce the reported result?

The remainder of the section studies five system responses to these pressures: serving-system design, architecture-level deployment design, streaming inference, efficiency optimization, and full-duplex convergence. These responses are not mutually exclusive. A single Speech LLM service may combine a deployment-aware model, a streaming policy, cache and decoding optimizations, a serving stack, and a full-duplex interaction loop.

Table 4 turns this structure into a reporting checklist. The first two columns identify the deployment pressure behind each failure mode. The third column links that pressure to the design patterns most likely to affect it, and the final column specifies the conditions that a deployment card should record. The table is not a leaderboard and does not enumerate deployment papers; it defines the information required to interpret deployment claims in the systems discussed below.

The corresponding subsections discuss the quantitative signals reported by these systems. We do not aggregate them into a leaderboard because the systems differ in task, hardware, latency definition, and interaction setting.

5.3. System-Level Serving

System-level serving determines whether Speech LLM inference can run as temporal-stream execution rather than isolated prompt processing. The central design question is how to coordinate input ingestion, acoustic encoding, language generation, speech synthesis, customization, and cluster scheduling under one serving loop while controlling latency fragmentation, memory growth, hardware mismatch, and interaction incompleteness.

VoxServe (Kamahori et al., 2026) introduces a serving system designed specifically for Speech Language Models [46]. Its main architectural idea is a model-execution abstraction that separates model architecture from system-level optimization. Modern SpeechLMs vary widely in their internal structures, including encoder-decoder ASR models, autoregressive speech generators, and codec-based synthesizers. Nevertheless, they share computational patterns such as attention over temporal sequences, streaming I/O, and multi-stage processing. VoxServe exploits

Failure mode	System pressure	Relevant design pattern	Deployment criteria
Latency fragmentation	First response, incremental output, and complete response measure different behaviors	Serving system; streaming inference	Report TTFB, token delay, end-to-end latency, streaming policy, and interaction-loop latency together.
Streaming mismatch	Complete-utterance models may behave differently under partial input	Streaming inference; architecture design	Separate offline and streaming settings; specify input chunk size, output granularity, and emission policy.
Memory growth	Long audio and video streams expand KV-cache, feature cache, and multimodal state	Efficiency optimization; serving system	Report input duration, output duration, cache policy, peak memory, and active compression or eviction methods.
Hardware mismatch	FLOPs and parameter count do not determine device-level behavior	Architecture design; serving system	Report device class, runtime stack, batch size, measured latency, measured memory, and edge/cloud placement.
Interaction incompleteness	Recognition or generation latency alone does not capture simultaneous listening and speaking	Full-duplex convergence; serving system	Evaluate simultaneous input/output, turn-taking, barge-in, interruption recovery latency, and response quality together.

Table 4
A deployment analysis framework for Speech LLMs. Each failure mode induces a system pressure, points to relevant design patterns, and implies concrete criteria for deployment-card reporting.

these shared patterns to support streaming-aware scheduling and an asynchronous inference pipeline, reporting 10–20x higher throughput than existing implementations at similar latency across several modern SpeechLMs.

NIM4-ASR (Xie et al., 2026) studies serving from the perspective of model-system co-design [106]. Instead of asking how to serve an arbitrarily large model efficiently, it asks what model size is sufficient for production-grade quality and how such a model should be served at scale. The resulting system uses a 2.3B-parameter encoder-LLM architecture with a clear division of labor: the encoder handles acoustic processing, while the LLM handles linguistic reasoning. The production system supports real-time streaming inference and million-scale hotword customization through retrieval-augmented generation (RAG) with sub-millisecond retrieval latency.

EPD Disaggregation (Singh et al., 2025) extends the prefill-decode disaggregation paradigm from text LLMs to multimodal serving [90]. Its main observation is that multimodal models introduce a separate bottleneck: encoding multimedia inputs such as video frames and audio spectrograms is expensive and operates on a different timescale from prefilling or token generation. The system separates serving into three independently scheduled phases, Encode, Prefill, and Decode, and adds a caching mechanism for multimedia tokens.

Evaluating Kubernetes Performance for GenAI Inference (Malleni et al., 2026) studies the infrastructure layer and reports substantial improvements in tail TTFT latency under Kubernetes-native routing and scheduling configurations, with the exact percentage depending on the metric and experimental setting [65]. Although this work is not specific to Speech LLMs, it is relevant because speech interaction is highly sensitive to tail latency.

The shared pattern is that serving decisions determine which part of the speech interaction loop becomes visible. VoxServe emphasizes heterogeneous temporal execution, NIM4-ASR emphasizes the production ASR boundary between acoustic encoding and linguistic reasoning, EPD Disaggregation emphasizes phase separation, and Kubernetes-level work emphasizes cluster behavior. Speech LLM deployment needs serving stacks that expose throughput, TTFT, and interaction-level traces under the same setup, including simultaneous input ingestion, output generation, interruption handling, cache growth, and tail latency under realistic audio durations.

5.4. Architecture-Level Deployment Design

Architecture-level work brings deployment constraints into the model design stage. Instead of treating latency, memory, and hardware behavior only as serving issues, these systems make them part of the architecture itself. This premise is especially important for edge and consumer hardware, where memory and latency limits cannot be addressed only through better serving.

LFM2 (Amini et al., 2025) uses hardware-in-the-loop architecture search under explicit edge latency and memory constraints [4]. The resulting hybrid backbone combines gated short convolutions with grouped query attention blocks and is optimized for measured performance on target hardware rather than theoretical FLOPs. The model family ranges from 350M to 8.3B parameters, includes dense and MoE variants, supports 32K context length, and is trained on 10–12 trillion tokens. The LFM2-Audio variant supports real-time speech-to-speech interaction, reports competitive quality relative to substantially larger models under its evaluation setting, and achieves 2x faster prefill and decode on CPUs than similarly sized models.

MiniCPM-o 4.5 (Cui et al., 2026) focuses on full-duplex interaction [21]. Many multimodal models still operate in alternating phases: they perceive and then respond. This design prevents the model from incorporating new inputs during generation or acting proactively. MiniCPM-o introduces Omni-Flow, a streaming framework that aligns omni-modal inputs and outputs on a shared temporal axis and transforms turn-based interaction into continuous, time-aligned processing. With 9B parameters and less than 12GB RAM, it is presented as an open-source model that supports simultaneous seeing, listening, and speaking with proactive behavior on consumer edge hardware.

S2M3 (Yoon et al., 2025) takes a modular approach to edge deployment by decomposing multi-modal models into reusable modules for distributed multi-task inference [114]. By allowing the same encoder modules to serve multiple downstream tasks, S2M3 reduces memory use by up to 62% and improves latency by 56.9% relative to cloud-based approaches. TinyM2Net-V3 (Rashid & Mohsenin, 2024) studies a more constrained setting, combining knowledge distillation with low-bit-width quantization for multimodal audio-visual inference under severe memory limits [86].

These systems represent four distinct deployment philosophies. LFM2 optimizes directly for measured hardware behavior, MiniCPM-o optimizes for interaction continuity, S2M3 optimizes for modular resource sharing, and TinyM2Net-V3 targets severe memory constraints through compression. Together, they show that deployment-aware architecture design spans trade-offs among latency, memory, modularity, and interaction capability. Speech LLM deployment needs architecture specifications that include device class, measured latency, memory footprint, supported context length, input audio duration, and output audio duration.

5.5. Streaming and Real-Time Inference

Streaming work makes output timing part of the inference objective. A streaming model must decide not only what to output, but also when to output it. Component-level latency and conversational-loop latency are related but not identical: a correct response that arrives too late may still fail the interaction.

Ultra-Low Latency End-to-End Streaming Speech Synthesis (Su et al., 2026) reports 48.99ms average TTFB and 10.6x acceleration over conventional cascaded pipelines under its evaluation setting [94]. The key idea is to remove the vocoder bottleneck by directly modeling the compressed discrete latent space of the Mimi neural audio codec. A modified FastSpeech 2 backbone uses progressive depth-wise sequential decoding across 32 residual vector quantization layers to generate speech in non-autoregressive blocks. Experiments in English and Malay provide cross-language evidence within the scope of the evaluated languages.

SpeakStream (Bai et al., 2025) addresses the LLM-then-TTS cascade used in many conversational AI systems [10]. Traditional TTS models are trained on complete utterances and can introduce large delays when paired with streaming LLM text. SpeakStream uses a decoder-only architecture trained with next-step prediction on interleaved text-speech data, enabling it to generate audio incrementally while receiving streaming text.

Streaming ASR with Decoder-Only LLMs (Wan et al., 2026) studies when an LLM-based ASR system should emit tokens relative to the input speech [101]. A read/write policy network with monotonic chunkwise attention reduces average token generation delay by 62.5% under the reported setup. Mini-Omni (Xie & Wu, 2024) was an early end-to-end system for real-time speech interaction without an external TTS system [105]. Its text-instructed speech generation method, batch-parallel inference strategy, and "Any Model Can Talk" training method showed that a single model could handle the complete speech interaction loop while preserving core language capabilities.

The shared pattern is that different systems optimize different latency boundaries. TTFB, first-token latency, token generation delay, and end-to-end latency answer different questions. A speech synthesis result with strong TTFB and an ASR result with low token delay provide complementary views of latency, but neither metric alone characterizes the full interaction loop. Speech LLM deployment needs latency optimization at the conversational-loop level, where component latency links to streaming policy, input chunk size, output granularity, and downstream user-visible response time.

5.6. Efficiency Optimization

Efficiency optimization treats decoding, compression, and cache management as deployment levers that can control memory growth and hardware mismatch without redesigning the entire model or serving stack. Real systems may combine quantization, speculative decoding, cache pruning, batching, and streaming policies, so the important deployment question is how these gains interact when they are composed.

5.6.1. Compression and Quantization

Quantizing Whisper-small (Sohler et al., 2026) evaluates post-training quantization across libraries and bit-widths for Whisper and provides practical guidance for constrained hardware [91]. Edge-ASR (Feng et al., 2025) expands this analysis to eight post-training quantization (PTQ) methods across Whisper and Moonshine, and identifies best practices for low-bit edge deployment [28]. Resource-Efficient Speech Quality Prediction (Nilsson et al., 2024) shows that binary activation maps with 8-bit quantization-aware training (QAT) can reduce inference memory by 25-fold for speech quality assessment [73]. These results suggest that speech models may tolerate aggressive quantization in some settings, but this tolerance depends on the task, architecture, and evaluation metric.

5.6.2. Speculative Decoding

Speculative decoding uses a fast draft model to propose tokens that a larger verifier accepts or rejects in parallel. SpecASR (Wei et al., 2025) reports 3.04–3.79x speedup over autoregressive decoding with no accuracy loss under its evaluation setting, using adaptive draft generation and a sparse token tree [103]. Accelerating Autoregressive Speech Synthesis (Lin et al., 2025) applies speculative decoding to TTS and reports a 1.4x speedup while maintaining fidelity and naturalness [56]. The smaller gain relative to ASR is consistent with the higher token entropy of speech synthesis.

Edge-Cloud Collaborative Speech Emotion Captioning (Xue et al., 2026) introduces Uncertainty-Guided Speculative Decoding for privacy-preserving edge-cloud collaboration [108]. A lightweight edge model drafts emotion captions locally, and only high-uncertainty token blocks are sent to a cloud verifier. Model-free Speculative Decoding (Ho et al., 2025) removes the draft model and instead uses precomputed n-gram token maps, reporting a 10% absolute speed improvement on CPU with minimal additional memory [35].

5.6.3. KV-Cache Optimization

AccKV (Jiang et al., 2025) addresses KV-cache growth in audio-video LLMs, where both modalities create long temporal sequences [43]. The key observation is that attention in higher transformer layers shifts toward the video modality regardless of task type. AccKV uses Layer Adaptive Focusing to identify important tokens by layer and modality, and Cross-Calibration to align low-priority modality caches with high-priority ones before selective eviction.

Across these studies, the shared pattern is modular efficiency: each method improves one pressure point in the deployment stack. Speech LLM deployment needs to treat these methods as composable system policies rather than isolated speedups. For example, evaluations can combine quantization, speculative decoding, and cache pruning under the same streaming input regime. This is the deployment analogue of evaluating individual benchmark capabilities together rather than only in isolation.

5.7. Deployment Evidence Under Realistic Pressure

Deployment evidence should specify whether a result comes from a product demonstration, a reproducible component benchmark, or a controlled system measurement. GPT-4o-like products define compelling user-facing targets, while open systems make instrumentation, ablations, and controlled measurement easier. Researchers often study Whisper-derived ASR models, codec-based TTS pipelines, and selected open multimodal models because these systems support reproducible instrumentation. Interaction-first systems such as MiniCPM-o 4.5, OpenOmni, GPT-4o-Audio-style real-time systems, and Gemini Live-style products provide references for interaction quality, although access, instrumentation, and reproducibility differ across systems.

Speech LLM deployment needs to separate three kinds of evidence: product-level demonstrations, reproducible component benchmarks, and controlled system measurements. Product demonstrations define experience targets; component benchmarks isolate mechanisms; controlled system measurements compare deployment behavior. Treating these as complementary forms of evidence makes cross-system comparison clearer without using one form as a substitute for another.

5.8. Full-Duplex and Omni-Modal Convergence

Full-duplex omni-modal systems treat perception and generation as concurrent rather than alternating processes. Full-duplex interaction is therefore not merely faster streaming; it changes the interaction contract. A system must process incoming audio or visual streams while generating speech, maintain low latency in both directions, accept interruptions, and coordinate multiple modalities under memory and compute constraints.

OpenOmni (Luo et al., 2025) introduces a two-stage progressive multimodal alignment framework with 7B parameters [63]. A notable result is cross-modal transfer: training a pre-trained speech model on text-image tasks yields near-zero-shot generalization from vision to speech and outperforms models trained on explicit tri-modal datasets. A lightweight non-autoregressive decoder trained with direct preference optimization produces real-time emotional speech with sub-1-second latency and a 5x speedup over autoregressive methods. Together with MiniCPM-o 4.5, OpenOmni illustrates a broader trend toward omni-modal models in the 7–9B range that can run on consumer hardware with real-time responsiveness.

This convergence changes the deployment target. The field no longer only needs faster ASR, faster TTS, or cheaper serving; it needs an interaction loop in which perception, reasoning, and speech generation occur continuously. Speech LLM deployment should treat full-duplex interaction as a first-class target and measure turn-taking, barge-in, interruption recovery latency, memory growth, and content quality together because these properties determine whether a system remains responsive and coherent during overlapping input and output.

5.9. Deployment Recommendations

Speech LLM deployment should design, optimize, and describe systems as real-time interaction loops rather than as independent ASR, LLM, and TTS components. Speculative decoding, quantization, KV-cache pruning, streaming serving, and edge deployment all improve different pressure points, but their value is clearest when they are tied to the deployed interaction regime.

First, deployment should move from isolated optimization to composed optimization. When a system combines quantization, speculative decoding, cache pruning, batching, or streaming policies, the evaluation should treat these choices as one deployment configuration whose combined behavior determines latency, quality, and memory growth.

Second, deployment should make streaming state explicit. Input chunk size, output granularity, cache policy, audio duration, interruption setting, and simultaneous input-output status define the runtime state of a Speech LLM. These details connect component speedups to the actual interaction loop.

Third, deployment should align latency metrics with interaction conditions. TTFB, average token delay, end-to-end latency, and time-to-completion each characterize a different user-visible behavior: first response, incremental output, complete answer delivery, or interruption recovery.

A deployment card summarizes these recommendations as a compact disclosure of the operational conditions under which an evaluation measures a deployment claim. At minimum, it should include model size, hardware, batch size, streaming policy, input duration, output duration, active optimizations, latency metric, quality metric, baseline implementation, and whether the evaluation uses cloud, edge, streaming, or full-duplex settings. This practice makes cross-paper comparison more tractable even when tasks differ.

The broader recommendation is to report both component-level measurements and interaction-loop measurements. A low TTFB becomes more interpretable when the evaluation connects it to conversational responsiveness. Similarly, evaluations should tie ASR token delay to downstream response delay when the target application is an assistant or agent. For full-duplex systems, evaluation should measure turn-taking, interruption handling, latency, and content quality together. This is the deployment counterpart of multi-channel evaluation: the system must be correct, responsive, and stable at the same time.

These deployment choices also determine the safety surface of Speech LLMs. Systems that stream continuously, retain acoustic context, split computation across edge and cloud, or synthesize speech in real time expose new risks related to privacy, impersonation, provenance, and audio-specific attacks. Section 6 turns to these safety, privacy, and governance implications.

6. Safety, Privacy, and Governance

Speech Large Language Models (Speech LLMs) extend language-model interaction from text to spoken communication, but this shift also changes the safety and privacy problem. Speech is not simply text represented as audio. It can serve as a command channel, an acoustic signal, a marker of speaker identity, and a carrier of synthetic media. A spoken utterance may encode words, speaker characteristics, prosody, emotion, accent, background sounds, and signal-level artifacts. These properties make speech useful for natural interaction, but they also create attack and misuse pathways that are not captured by text-only safety evaluation.

This section reviews six groups of risks and mitigations: input-side audio jailbreaks and adversarial speech, audio safety benchmarks, guardrails and anti-cloning defenses, synthetic voice misuse, speaker privacy protection, and documentation for governance. Because many of the cited studies are recent preprints, their findings should be interpreted cautiously. Still, the literature points to a consistent pattern: safeguards developed for text-only LLMs do not necessarily transfer to speech-capable systems. Speech LLMs therefore require safety analysis that accounts for audio representations, speech generation, speaker identity, and deployment context.

6.1. Speech-Specific Risk Surface

Speech LLMs expose semantic, acoustic, biometric, and synthetic-media risks at the same time. In text-only LLMs, harmful prompts are usually represented as discrete text tokens. In Speech LLMs, the input may pass through raw waveform encoders, speech tokenizers, ASR modules, acoustic projectors, or multimodal alignment layers before reaching the language backbone. These processing stages can change model behavior and create attack opportunities that are difficult to detect from transcripts alone.

6.1.1. Input-Side Attacks: Harmful Spoken Prompts and Modality Transfer Failure

Red-teaming studies show that safety behavior can change when harmful requests are presented as speech rather than text. **Audio Is the Achilles' Heel** evaluates Qwen-Audio, Qwen2-Audio, SALMONN-7B, SALMONN-13B, and Gemini-1.5-Pro under three settings: harmful questions in text and audio formats, harmful text paired with distracting non-speech audio, and speech-specific jailbreaks. Beyond aggregate attack success, the paper analyzes representation-space behavior using t-SNE visualizations and response-opening patterns. The authors find that harmful questions can become more effective when moved into the audio modality for several open-source audio LMMs, and that non-speech audio can reshape representation spaces in ways that affect safety behavior [110]. This result suggests that safety alignment learned through text may weaken after multimodal adaptation.

Adversarial audio attacks further show that the waveform itself can be optimized to bypass safety constraints. **AdvWave** treats jailbreak attacks against large audio-language models as a waveform-level optimization problem. The method includes dual-phase optimization to address gradient shattering from audio encoder discretization, adaptive target search to handle response variability, and classifier-guided natural-sound constraints to keep perturbations perceptually plausible [47]. This matters because the attack can operate below the transcript level: the spoken request may remain understandable to humans, while the acoustic perturbation changes how the model processes or answers it.

Multimodal attacks add another path. **JAMA** proposes a joint audio-text attack against spoken language models by combining gradient-based text suffix optimization with audio perturbation. The method jointly optimizes across modalities and evaluates both fixed-prompt and prompt-adjusted settings. Experiments show that combined text-audio attacks can outperform unimodal attacks across several spoken language models [49].

Sparse token attacks show that only parts of the audio sequence may be sufficient for a successful jailbreak. **TAGO** analyzes token-aligned gradients and finds that gradient energy is not evenly distributed across audio tokens. Its method identifies high-contribution token regions and optimizes only the corresponding waveform segments. Comparisons with full-waveform attacks, token-selection strategies, and efficiency analyses show that sparse token optimization can reduce attack cost while maintaining strong jailbreak performance [27]. This suggests that small local changes in audio should not be treated as automatically harmless.

Across these attacks, harmfulness is no longer only a property of the transcript. It can be introduced, strengthened, or hidden through the acoustic path. A speech-specific threat model should therefore include harmful spoken prompts, distracting non-speech audio, adversarial waveform perturbations, multimodal audio-text jailbreaks, speech-specific jailbreak prompts, and sparse token-level manipulation.

6.1.2. Output-Side Misuse: Synthetic Voice and Impersonation

Speech generation introduces a different class of risk. A generated voice can carry both content and an apparent speaker identity. Voice cloning is therefore more than harmful text generation with audio output; it can support impersonation, fraud, reputational harm, evidence manipulation, and social engineering. A speech assistant that can listen, reason, act, and reply in a realistic voice may improve accessibility and interaction, but it can also scale impersonation if voice generation is not constrained.

Voice cloning misuse can be grouped into three categories:

- **Identity impersonation:** a malicious actor uses a cloned voice to appear to be another person.
- **Evidentiary manipulation:** synthetic audio is presented as proof that someone said something.
- **Social engineering:** a cloned voice increases trust in scams, phishing, coercion, or fraudulent instructions.

Each category calls for different controls. Identity impersonation requires consent and anti-cloning protection; evidentiary manipulation requires provenance and forensic verification; social engineering requires organizational procedures such as callback verification, transaction limits, and human review.

6.1.3. Privacy Leakage: Speaker Identity and Paralinguistic Attributes

Speech privacy extends beyond the literal words being spoken. Even benign speech can reveal speaker identity, accent, age, emotional state, or background context. Speech LLMs that process raw or lightly processed audio may therefore collect more personal information than the user intended to provide. Stored audio and speaker embeddings can also become biometric records that later enable identity recovery or voice cloning.

This distinction separates two related risks. Voice cloning and deepfakes concern synthetic speech as an output misuse problem. Speaker privacy concerns risks that arise from the user's input speech, even when the content is harmless. Raw audio, voice embeddings, transcripts, and paralinguistic features can all contribute to identity leakage. Speech LLM systems should therefore treat speech as both conversational content and personal data.

6.2. Audio Safety Evaluation and Benchmarking

As audio jailbreak methods become more varied, benchmark design becomes central to safety evaluation. Early demonstrations often used small prompt sets or single-model case studies. Newer benchmarks aim to test attacks, models, defenses, perturbations, and evaluation protocols under more controlled settings.

6.2.1. Unified Jailbreak Benchmarks

JALMBench provides a broad framework for evaluating jailbreak vulnerabilities in large audio-language models. Its dataset includes harmful query samples, text-transferred jailbreak samples, and audio-originated jailbreak samples. It synthesizes audio variants with different accents, genders, TTS methods, and languages, and evaluates multiple LALMs across several attack and defense methods. The paper also reports evaluator reliability analysis, including repeated judging, cross-model consistency, and human consistency verification [82]. These details are important because benchmark results depend not only on the prompt set, but also on how harmfulness is judged.

The JALMBench results support several design lessons. Non-adversarial harmful queries can have different attack success rates in text and audio modalities. Stronger attacks such as AdvWave can achieve high attack success under the benchmark's settings. Topic sensitivity, attack efficiency, voice diversity, and architecture effects can all influence safety outcomes. These findings suggest that safety evaluation should report accent, speaker, and language variation explicitly, because robustness observed under one speech variety may not generalize to another [82].

6.2.2. Perturbation-Based Benchmarking

Audio Jailbreak provides a complementary benchmark focused on realistic speech and perturbation. Its base dataset converts adversarial text prompts into speech across policy-violating categories. The paper also introduces an Audio Perturbation Toolkit that applies transformations in the time, frequency, and amplitude domains while aiming to preserve semantic consistency. These transformations test whether safety behavior remains stable under speed changes, noise, clipping, pitch shifts, and related distortions [92]. This matters because deployed speech systems rarely receive clean, studio-quality audio.

Jailbreak-AudioBench combines a toolbox, dataset, and benchmark for studying audio jailbreaks. It introduces explicit and implicit jailbreak audio examples and evaluates multiple large audio-language models. One contribution is its focus on audio editing as a jailbreak mechanism: harmful semantics can be embedded or transformed through audio operations rather than simply read aloud as a standard prompt [17]. This is especially relevant for end-to-end Speech LLMs, which may preserve acoustic information that cascaded ASR-to-LLM systems discard.

6.2.3. Evaluation Requirements

Across these benchmark papers, several evaluation requirements emerge:

- Models should be tested on clean harmful spoken prompts.
- Models should be tested on perturbed harmful prompts.
- Benchmarks should include text-transferred and audio-originated jailbreaks.
- Evaluation should include multimodal audio-text attacks.

- Speaker, accent, language, and environmental variation should be reported.
- Defenses should be evaluated together with benign utility.

The benchmark gap is therefore not only about dataset size. Audio safety scores depend on perturbation choices, voice diversity, judge stability, target-model assumptions, and benign-utility trade-offs. Benchmark papers should report target models, judges, safety categories, attack assumptions, and whether evaluation relies on human review, model-based judging, or rule-based filtering.

6.3. Mitigation: Guardrails, Anti-Cloning, and Privacy Defenses

The attack and benchmark literature shows that text-only moderation is insufficient for Speech LLMs. If an attack is encoded in the acoustic signal, a filter applied only to the transcribed text may miss it. If a system removes too much acoustic information before model processing, it may also harm useful capabilities such as accent robustness, emotion understanding, speaker adaptation, or paralinguistic reasoning. Mitigation therefore has to cover jailbreak guardrails, anti-cloning defenses, privacy protection, and deployment monitoring.

6.3.1. Audio-Language Guardrails

ALMGuard provides one example of an audio-language-specific defense. The paper argues that defenses transferred from traditional audio adversarial attacks or text jailbreak defenses are limited against ALM-specific threats. The method identifies safety shortcuts in model activations and introduces safety shortcut activation perturbations as a runtime defense. It also uses a Mel-Gradient Sparse Mask to preserve benign utility. The paper evaluates the defense against multiple attack methods and includes analyses of benign task preservation [44]. This supports the need for guardrails that operate at the representation or activation level, not only at the transcript level.

Current guardrail research does not yet cover all parts of the threat model equally. Speech-specific jailbreak prompt transformations and sparse token-level manipulation remain only partially addressed by existing defenses. Future defenses may need to combine input detection, token-level robustness analysis, activation-level guardrails, and output monitoring.

6.3.2. Anti-Cloning Defenses

For speech synthesis and voice cloning, defenses must also address output-side misuse. **E2E-VGuard** focuses on unauthorized voice cloning in production LLM-based end-to-end speech synthesis. The paper's threat model describes API-based cloning workflows in which users upload only audio, while the platform internally performs ASR and TTS training. The method protects both timbre and pronunciation: timbre protection uses an encoder ensemble and MFCC-based feature loss to reduce speaker similarity in synthesized audio, while pronunciation protection attacks ASR transcription so that fine-tuning receives incorrect text-audio pairs. The paper also uses a psychoacoustic model to keep perturbations difficult for humans to perceive and evaluates across open-source synthesizers, commercial APIs, ASR systems, languages, and speakers [119]. This shows that anti-cloning defense has to consider the full speech synthesis pipeline, not only the final waveform.

6.3.3. User-Side Voice Privacy Protection

User-side privacy protection is another defense direction. **Enkidu** proposes real-time audio privacy protection against personalized voice deepfake threats using universal frequency-domain perturbations. The method does not assume white-box access to the attacker's cloning model. Instead, it uses black-box knowledge, few-shot training, and frequency-domain perturbations designed to preserve perceptual quality while reducing cloneability. The experiments evaluate privacy protection, audio quality, real-time efficiency, and robustness across settings [29]. This is relevant for users who want to publish or transmit speech while reducing the risk that their voice can later be cloned.

6.3.4. Adaptive De-Anonymization and Purification Attacks

Privacy defenses should be evaluated against adaptive attackers. **VoxATtack** studies a multimodal attack against voice anonymization systems. Its key insight is that anonymized speech may still leak identity when acoustic and textual information are combined. The method extracts both acoustic embeddings and text embeddings, then fuses them for de-anonymization. Experiments against VoicePrivacy benchmark settings show that multimodal attacks can outperform acoustic-only baselines [3]. For Speech LLMs, this is important because these systems often process both audio and text representations. Even if the voice is transformed, the transcript may contain linguistic habits, topic choices, names, or discourse patterns that help re-identify the speaker.

VocalBridge further challenges perturbation-based privacy defenses. The paper studies an adaptive attacker who purifies protected speech before using it for voice cloning or speaker verification. Its method uses a diffusion-bridge model in the EnCodec latent space to map protected speech back toward clean speech, with an optional Whisper-guided phoneme conditioning variant. The paper defines user objectives such as imperceptibility, verifiability, and inimitability, and attacker objectives such as restoring verifiability and breaking inimitability. Its evaluation across proactive defenses, voice synthesis tools, and speaker verification systems suggests that perturbation-based protections may be fragile against purification [1].

6.3.5. Layered Defense Architecture

A practical mitigation architecture for Speech LLMs should include several layers:

- **Input-level defenses:** inspect audio for suspicious transformations, hidden instructions, adversarial perturbations, or abnormal acoustic patterns.
- **Representation-level defenses:** monitor acoustic embeddings or hidden states for attack-like behavior.
- **Transcription-level defenses:** inspect ASR output while recognizing that transcription is not sufficient on its own.
- **Output-level defenses:** moderate generated text and generated speech before delivery.
- **Privacy controls:** minimize raw audio storage, separate content processing from identity processing, and document whether speaker embeddings are stored.
- **Deployment-level controls:** include logging, rate limits, human escalation, and incident response for high-risk applications.

These defenses shift Speech LLM safety from a single-filter problem to a layered systems problem. Because attacks may target the waveform, transcript, internal representation, generated speech, or deployment workflow, defenses should be tested under adaptive conditions.

6.4. Identity Misuse and Synthetic Speech Provenance

Synthetic speech risks are related to privacy risks, but they are not the same. The previous subsection focused on defenses against jailbreaks, cloning, and identity leakage. This subsection focuses on generated speech after it leaves the model: whether it can be detected, attributed, and traced.

6.4.1. Deepfake Detection Under Realistic Conditions

The **Perturbed Public Voices ($\mathbf{P^2V}$)** dataset examines the weakness of audio deepfake detectors under realistic conditions. Its dataset includes identity-consistent transcripts, environmental and adversarial noise, and voice cloning systems from multiple years. The paper evaluates several detectors and shows that performance drops under stronger perturbations and newer cloning methods [30]. This supports the argument that deepfake detection should not be validated only on clean or closed-set benchmarks. Speech LLM governance should assume that generated or cloned audio may be compressed, re-recorded, mixed with background noise, or produced by newer synthesis systems than those seen during detector training.

6.4.2. Robustness Limits of Audio Watermarking

Watermarking and provenance mechanisms help identify AI-generated speech and support misuse investigation. However, audio can be compressed, resampled, re-recorded, denoised, edited, cropped, transformed by neural codecs, or regenerated by another model. Each operation can weaken or remove a watermark.

A systematic study of audio watermark robustness surveys 22 watermarking schemes and evaluates removal attacks across signal-level, physical-level, and AI-induced distortions. The paper categorizes watermarking approaches, builds an evaluation pipeline, and tests robustness under many removal configurations. The authors conclude that no surveyed scheme is robust against all tested distortions in practice [104]. This is a caution for Speech LLM deployment: watermarking can support provenance, but it cannot be the only safeguard.

Deep Audio Watermarks are Shallow focuses on limitations of post-hoc speech watermarking. The paper analyzes how post-hoc methods manipulate low-level audio features after generation and shows that transformation-based removal can reduce detectability while preserving audio quality. Its evaluation shows why watermark robustness depends on the attacker's ability to process or transform the signal without knowing the watermarking scheme [75]. This suggests that post-hoc watermarking should be supplemented by generation-time provenance, secure metadata, or model-integrated methods.

6.4.3. Generation-Time and Dual-Provenance Watermarking

Several newer methods propose stronger watermarking strategies. **MelShield** embeds keyed payloads into Mel-spectrogram representations during TTS generation. Its method integrates watermarking into the TTS pipeline rather than relying only on post-processing, and its experiments evaluate attribution accuracy, robustness, and audio quality. This is useful for Speech LLMs because speech generation systems often operate through intermediate acoustic representations such as Mel-spectrograms, making internal watermarking more natural than waveform-only post-processing [45].

DualMark extends provenance from model-level attribution to both model and training-data attribution. The paper defines a dual-level attribution problem, embeds model and data watermarks into Mel-spectrogram representations during training, and uses watermark consistency loss so generated audio preserves decodable provenance. Its experiments compare model-level and data-level attribution, ablate the watermark consistency loss, and evaluate robustness against pruning, compression, resampling, and noise [112]. This is relevant because accountability may require identifying both the generating model and the data source.

AWARE proposes watermarking through adversarial resistance to edits. Its method is designed around the possibility that watermarked audio may be edited, cropped, compressed, or desynchronized. Instead of relying only on fixed simulated distortions, it trains the watermarking system with adversarial resistance and uses a detector that aggregates temporal evidence. This direction is important because synthetic speech may be transformed many times before detection [80].

The watermarking literature reframes provenance as a robustness-under-transformation problem. The question is not only whether a watermark can be detected on clean generated audio, but whether it survives compression, re-recording, editing, adversarial removal, and model-based transformation.

6.4.4. Traceability Stack

For Speech LLM deployment, provenance should be treated as a stack rather than a single watermark. A traceability system may include:

- Audio watermarking.
- Secure metadata.
- Model version records.
- Consent records.
- User identity verification for voice cloning.
- Audit logs and incident response procedures.

In public-facing or high-risk settings, generated speech should remain traceable even when the audio watermark fails. Watermarking is useful but insufficient: it supports accountability, but it has to be combined with metadata, policy, monitoring, and operational safeguards.

6.5. Governance, Documentation, and Compliance Gaps

The traceability stack above describes technical provenance mechanisms. This subsection turns to documentation and governance. Speech LLMs need documentation that covers not only model accuracy, but also intended use, prohibited use, data provenance, speaker privacy, audio safety testing, watermarking, logging, and incident response. General AI governance frameworks provide useful starting points, but they need speech-specific extensions.

6.5.1. Model Documentation

Model Cards offer a starting point for standardized model reporting. The paper proposes short documents accompanying trained models, including intended use, performance evaluation, ethical considerations, and performance across relevant groups and conditions. Although the original framework focuses on machine learning models more broadly, the same structure can be adapted to Speech LLMs [68]. A Speech LLM model card should report supported languages and accents, evaluated noise conditions, speech input and output capabilities, audio jailbreak testing, deepfake risk controls, privacy protections, latency constraints, and whether raw audio or speaker embeddings are stored.

6.5.2. Machine-Readable Governance

Machine-readable documentation may help connect policy requirements to system behavior. **AI Cards** propose human- and machine-readable documentation for AI systems, including technical specifications, context of use, and risk management information inspired by EU AI Act requirements. This type of structured documentation is useful for Speech LLMs because deployment risk depends heavily on context: a speech model used for entertainment has different requirements from one used in healthcare, finance, education, or public services [32].

Policy Cards further extend this idea to runtime governance. They encode operational, regulatory, and ethical constraints as machine-readable artifacts for autonomous AI agents and emphasize auditability, compliance, transparency, and runtime policy enforcement [66]. For speech assistants that can call tools or act in real time, runtime governance can help translate high-level rules into operational controls.

6.5.3. Governance Requirements for Speech LLMs

A governance framework for Speech LLMs should include at least five components:

- **Data governance:** require consent and provenance documentation for speech data, especially when voices are identifiable or cloneable.
- **Safety governance:** require audio-specific red teaming, including clean harmful speech, perturbed audio, hidden-semantics attacks, and multimodal audio-text jailbreaks.
- **Privacy governance:** restrict raw audio retention, speaker embedding storage, and sensitive inference from paralinguistic cues.
- **Provenance governance:** require synthetic speech disclosure, watermarking where feasible, and secure audit trails.
- **Operational governance:** define escalation paths, human review, incident reporting, and post-deployment monitoring.

Most existing documentation frameworks are modality-general, while Speech LLMs create speech-specific risks. Voice is an interaction medium, a biometric signal, and a synthetic media carrier. Governance documentation should therefore specify how the system handles audio data, speaker identity, generated speech, and misuse reports.

6.6. Design Implications

The reviewed literature suggests several design implications for Speech LLM development and deployment.

- **Data collection:** systems should document consent, voice provenance, and speaker privacy risks.
- **Training and adaptation:** developers should examine whether multimodal alignment changes safety behavior inherited from the text backbone.
- **Evaluation:** benchmarks should include audio jailbreaks, perturbation robustness, accent and language variation, privacy leakage, and deepfake misuse.
- **Deployment:** systems should use layered guardrails, provenance mechanisms, logging, and human escalation.
- **Post-deployment monitoring:** organizations should track misuse patterns, update defenses, and disclose incidents where appropriate.

Table 5
Coverage comparison: this survey versus five recent Speech LLM and audio-language reviews. ✓ indicates substantive coverage; – indicates limited or absent coverage. Shaded cells highlight coverage unique to this survey.

Survey	Training	Meta-evaluation of benchmarks	Deployment / serving	Safety / privacy	Lifecycle coupling	Methodology type
Cui et al. 2025 [20]	✓	–	–	–	–	Narrative
Peng et al. 2025 [81]	✓	Capability taxonomy only	–	–	–	Narrative
Ji et al. (WavChat) [42]	Partial	–	Dialogue / duplex only	–	–	Narrative
Latif et al. 2023 [51]	✓ (speech foundation models)	–	–	–	–	Narrative
Ye et al. 2024 (LVLM safety) [113]	–	–	–	✓ (vision-centric)	–	Narrative
This survey	✓	✓ (validity-aware)	✓ (interaction-oriented)	✓ (speech-specific)	✓	Lifecycle audit

The central design point is that speech is not just text with sound. It is a high-dimensional signal that can carry semantic content, speaker identity, emotion, and contextual information. Audio jailbreaks attack speech as a command channel. Adversarial perturbations attack speech as a signal. Voice cloning attacks speech as identity. Deepfakes attack speech as evidence. Privacy attacks exploit speech as personal data. Watermarking and provenance govern speech as synthetic media. Speech LLM safety therefore depends on technical robustness, privacy protection, provenance, and operational controls. Section 7 discusses open problems in audio-specific red teaming, adaptive defenses, privacy-preserving speech representations, and provenance under audio transformation.

7. Future Directions

Table 5 clarifies the contribution of this survey relative to prior work by comparing its scope with five recent reviews of Speech LLMs and audio-language models.

The remainder of this section adopts a lifecycle perspective to organize future directions around stage-specific challenges and cross-stage dependencies that shape practical Speech LLM systems.

Training Closing the data imbalance for low-resource languages and non-Latin scripts will require automated data generation, cross-lingual transfer, and quality-aware weakly supervised pipelines. Alignment modules need to preserve paralinguistic content while remaining streaming-friendly, and preference alignment needs reward signals that capture acoustic and prosodic quality, not only transcript fidelity.

Evaluation Benchmarks should adopt joint evaluation across content fidelity, paralinguistic awareness, interaction dynamics, latency, and safety; distribution-aware metrics for subjective targets; mandatory human baselines for new capability claims; and standardized Speech-LLM Evaluation Cards for cross-paper comparability (Section 4).

Deployment Compositional optimization—combining quantization, speculative decoding, KV-cache reduction, and streaming serving in a single full-duplex stack—remains an open engineering problem. Reporting standards should include latency under simultaneous input and output, not only offline batch throughput (Section 5).

Safety, privacy, and governance Audio-specific red teaming, voice anonymization that resists adaptive purification, watermarking that survives realistic transformations, and modality-aware governance documentation are all immediate priorities (Section 6).

Cross-stage coupling The most impactful research questions sit between stages: training pipelines that anticipate streaming deployment, benchmarks that score safety and capability jointly, and governance frameworks that translate technical risk into deployable controls. A lifecycle perspective, rather than further specialization within a single stage, is likely to produce the largest gains for real-world Speech LLM systems.

7.1. Limitations

Several limitations qualify this survey. First, the benchmark corpus analyzed in Section 4 is a 2024–2026 snapshot; fifteen of the eighteen benchmarks were arXiv preprints at the time of writing and may have been peer-reviewed or superseded since. Claims that depend on this corpus should therefore be read as time-bounded. Second, the capability coding behind Figure 3 was performed by a single reviewer; we have released the per-benchmark notes as supplementary material to enable community validation, but we do not report formal inter-rater agreement. Third, the coverage analysis is biased toward English-language and Anglophone-dominated benchmarks, which under-represents code-switching, dialect, and non-Latin script regimes. Fourth, the deployment and safety syntheses draw heavily on system descriptions from authors of those systems, which limits adversarial scrutiny of reported latency and security measurements. Fifth, the proposed Evaluation Card and Deployment Card templates are not yet empirically validated against community use; they are framing tools for reviewer consensus, not measurement instruments. We acknowledge these limitations to preempt over-generalization of the lifecycle audit and to invite community extension of the supplementary material.

8. Conclusion

This survey reviewed Speech Large Language Models by connecting training, evaluation, deployment, and speech-specific risk analysis. Training advances such as Whisper-style weakly supervised pretraining, alignment modules including AlignFormer and Q-Former, parameter-efficient fine-tuning, and preference alignment via DPO and RLAIF have made Speech LLMs increasingly capable. Evaluation, however, remains a bottleneck: current benchmarks over-measure inherited ASR and emotion tasks while under-measuring latency, safety, code-switching, and privacy. Deployment research has begun to address streaming, edge, and full-duplex constraints, yet integrated systems combining serving, architecture, and algorithmic optimization remain rare. Safety, privacy, and governance work has shown that text-only safeguards do not transfer to speech: audio jailbreaks, voice cloning, deepfake misuse, and speaker re-identification require modality-specific responses. By treating Speech LLMs as systems rather than only models, the lifecycle perspective surfaces both the structural gaps and the cross-stage couplings that will shape the next generation of robust, efficient, and responsible spoken-language systems.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work, the authors used generative AI tools to assist with language polishing, structural revision, and checklist-based editing. The authors reviewed, verified, and edited all content as needed and take full responsibility for the content of the submitted manuscript.

Data Availability

The benchmark coding matrix and supplementary reading notes supporting this survey are provided as supplementary material.

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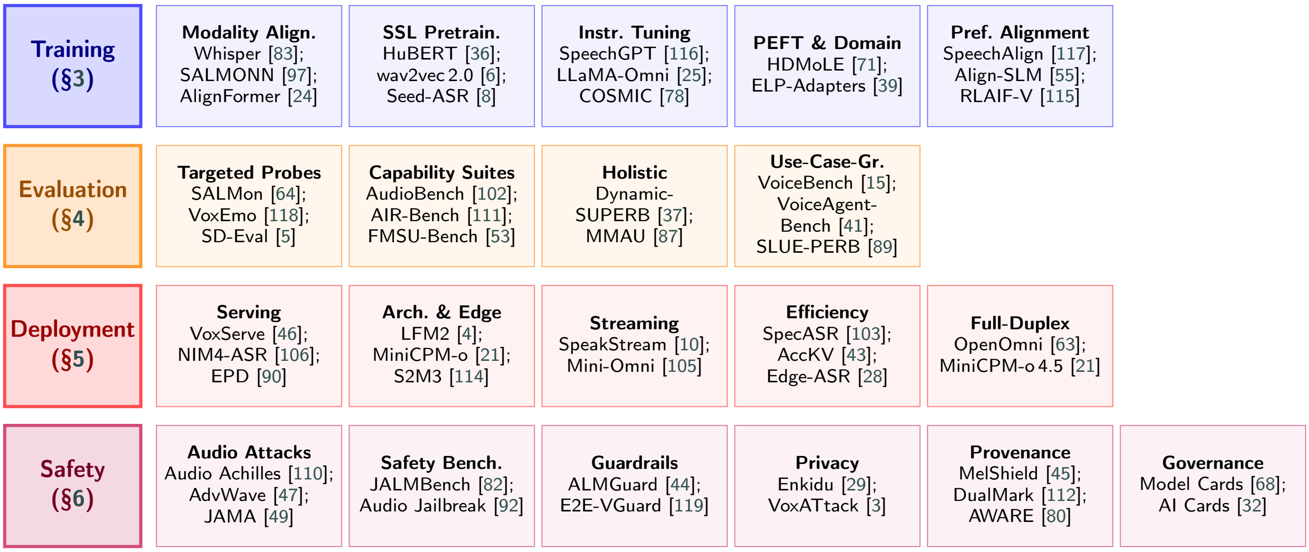

Figure 1: Paper taxonomy of Speech LLM lifecycle research covered by this survey. Representative methods and systems are grouped under the four lifecycle stages introduced in Figure 2. The taxonomy lists 2–3 representative works per sub-topic to indicate scope; the full reference list and benchmark coding are released as supplementary material.

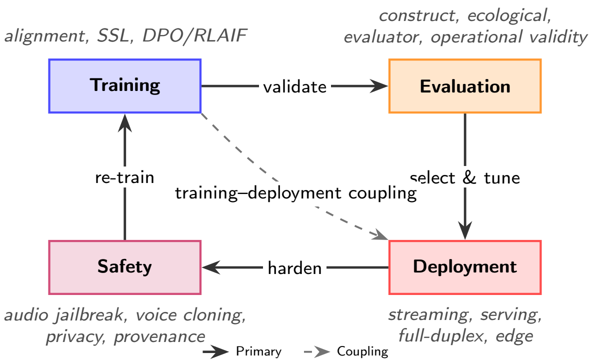

Figure 2: The Speech LLM lifecycle as a coupled system. Solid arrows trace the conventional cycle from training through evaluation, deployment, and safety, with re-training closing the loop. The dashed diagonal highlights the survey's central claim that the cycle is not purely sequential: training choices shape deployment feasibility, and the consequences of evaluation, deployment, and safety choices feed back through re-training.

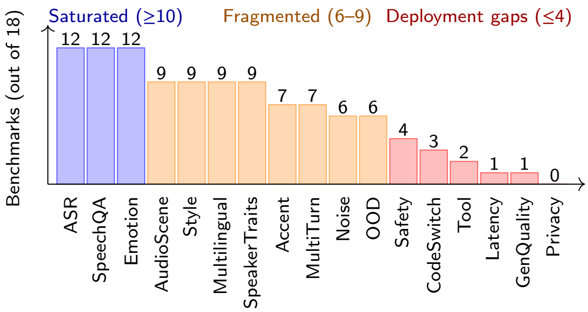

Figure 3: Capability coverage across 18 Speech LLM benchmarks (2024–2026). Each bar shows how many benchmarks in our corpus explicitly evaluate the capability. Coverage is dominated by ASR, spoken question answering, and emotion; deployment-facing dimensions (safety, code-switching, tool use, latency, generation quality, privacy) remain weakly covered. The underlying 18×17 coverage matrix is released as supplementary material.